

Demand Modeling in Urban Air Mobility

Chiu Yu KO

Low Altitude Economics

Week 4

Learning Objectives

By the end of this class, you should be able to:

- **Build** a simple demand forecast from potential trips, corridor share, mode share, and operating constraints.
- **Estimate** willingness to pay using generalized cost and interpret a simple demand curve.
- **Explain** how own-price elasticity and customer heterogeneity affect pricing decisions.
- **Incorporate** public acceptance as a transparent scenario assumption rather than a fixed universal multiplier.
- **Stress-test** demand and revenue under weather, capacity, and policy scenarios.

Would Customers Actually Choose It?

Imagine a proposed airport air-mobility service in Hong Kong.

- The aircraft may be technically capable of completing the trip quickly.
- Customers still compare the complete journey with rail, taxi, bus, and private car alternatives.

The commercial question is not simply whether the service can fly. It is whether enough customers will choose it at a viable price.

- **Quick poll (2 min):** Which factor is likely to matter most: price, door-to-door time, reliability, convenience, or public confidence?

Concept Primer: WTP and Demand Curves

Willingness to pay (WTP) is the maximum amount a particular customer would pay for a service rather than go without it or choose an alternative.

- Customers have different WTP because they value time, reliability, comfort, and urgency differently.
- A customer does not necessarily disappear from the entire market when price exceeds WTP; they may choose another mode.

A demand curve shows the quantity customers are expected to purchase at different prices, holding other important conditions fixed.

- It helps managers compare volume, revenue, capacity requirements, and contribution at alternative prices.

Applied Tool: A Simple Linear Demand Curve

The Formula

$$Q = a - bP$$

- Q = expected quantity demanded during a defined period
- P = price per unit
- a = estimated demand at a zero price under the model
- b = change in quantity associated with a HK\$1 price increase
- The choke price is $P = a/b$, where estimated demand reaches zero.

Managerial caution: A survey does not automatically produce a reliable demand curve. Managers should use multiple observations, realistic alternatives, and validation against actual behaviour where possible.

Back-of-the-Envelope: Comparing Prices for an Airport Service

Illustrative demand model: $Q = 8,000 - 8P$. Demand falls by 800 passenger-trips for each HK\$100 price increase.

Price = HK\$300

- $Q = 8,000 - 8(300) = 5,600$
- Revenue = $300 \times 5,600$
- **Revenue = HK\$1.68 million**

Price = HK\$500

- $Q = 8,000 - 8(500) = 4,000$
- Revenue = $500 \times 4,000$
- **Revenue = HK\$2.00 million**

Takeaway: HK\$500 maximizes revenue for this particular linear model. A profit-maximizing price would also require cost and capacity information.

Illustrative assumptions for classroom analysis; not a survey result or market forecast.

Concept Primer: What is Demand Modelling?

Demand modelling uses evidence and explicit assumptions to estimate how many customers may choose a service under different conditions.

- It is not a precise prediction of exactly how many people will travel.
- It combines market size, trip patterns, competing modes, price, time, reliability, access, and customer preferences.
- The forecast should be presented as a range or set of scenarios rather than a single certain number.

Link to Week 3: Does forecast demand fit the price, capacity, and activity required to break even?

A Four-Step Travel-Demand Framework

A traditional transport-planning framework can organize an LAE demand forecast:

1. **Trip generation:** How many relevant trips may occur?
2. **Trip distribution:** Which origins and destinations do those trips connect?
3. **Mode choice:** What share may choose LAE rather than rail, taxi, bus, or car?
4. **Network assignment:** How would the chosen trips use the available network, routes, and facilities?

NASA-sponsored UAM studies use detailed demand and market models, but the four-step structure is a general transport-planning framework rather than a unique “NASA formula.”

Applied Tool: A Simple Demand-Funnel Approximation

A Managerial Screening Formula

Potential LAE Passenger-Trips = Relevant Trips $\times$ Corridor Share $\times$ Mode Share

Capacity Check

Served Passenger-Trips = $\min\{\text{Potential Demand}, \text{Available Capacity}\}$

- The multiplication is a simplified demand funnel, not the full four-step transport model.
- Route assignment allocates trips across a network; it should not normally be treated as an arbitrary demand-reduction percentage.
- Keep passenger-trips, flights, and seats separate throughout the model.

Back-of-the-Envelope: Demand and Capacity for an Airport Corridor

Assumptions: 20,000 relevant daily trips; 60% involve the selected corridor; estimated LAE mode share = 35%.

Potential Demand

- $20,000 \times 0.60 \times 0.35$
- **4,200 passenger-trips/day**
- This is customer demand before checking operating capacity.

Capacity-Constrained Service

- Assume 300 flights/day
- Four seats and 80% target load factor
- Capacity = $300 \times 4 \times 0.80$
- **960 passenger-trips/day**

Takeaway: Potential demand may exceed available capacity. The business model must distinguish customers who want to travel from customers the system can actually serve.

Illustrative assumptions for classroom analysis; not current corridor data.

Concept Primer: Generalized Cost

Generalized cost combines the main burdens of a complete journey into a common monetary measure.

- fare and related charges;
- access, waiting, boarding, in-vehicle, and transfer time;
- reliability, inconvenience, and other service attributes where these can be estimated credibly.

Managerial use: Compare door-to-door alternatives on a consistent basis. *A lower generalized cost increases the attractiveness of a mode, but does not imply that every customer will choose it. Preferences differ and forecasts remain probabilistic.*

Applied Tool: Generalized Cost

A Simple Formula

$$GC = \text{Fare} + VOT \times \text{Door-to-Door Time} + \text{Other Monetized Burdens}$$

- *VOT* converts travel time into money for comparison.
- Other burdens may include unreliability, difficult access, transfers, or screening, provided the values are estimated transparently.
- Avoid double counting. A weather delay should not appear in both expected travel time and a separate penalty unless the terms capture different effects.
- Customers are more likely, not certain, to select alternatives with lower generalized cost.

Back-of-the-Envelope: Comparing Door-to-Door Generalized Cost

Assumption: VOT = HK\$15 per minute. Other monetized burdens are illustrative.

Ground Taxi

- Fare: HK\$350
- Door-to-door time: 45 minutes
- Other burden: HK\$50
- $GC = 350 + 15(45) + 50$
- **GC = HK\$1,075**

Air-Mobility Service

- Fare: HK\$800
- Door-to-door time: 12 minutes
- Other burden: HK\$80
- $GC = 800 + 15(12) + 80$
- **GC = HK\$1,060**

Takeaway: The air service is slightly more attractive under these assumptions. The difference is small, so changes in waiting time, reliability, or perceived risk could reverse the result.

Illustrative assumptions for classroom analysis.

Session 2

WTP Estimation, Elasticities & Heterogeneity

Concept Primer: "Elasticity of Demand"

What happens when you change your ticket price?

- If you raise the price of a flying taxi by 10%, will you lose 5% of your customers, or 50% of your customers?
- **Elasticity** measures this exact sensitivity.

Two Types of Elasticity:

- **Own-Price Elasticity:** How your demand changes when *you* change your price.
- **Cross-Elasticity:** How your demand changes when *your competitor* changes their price. (e.g., If the MTR raises its fare by HK\$10, how many people suddenly switch to your flying taxi?)

Applied Tool: Price Elasticity of Demand

The Formula

$$E_d = \frac{\% \text{ change in quantity demanded}}{\% \text{ change in price}}$$

- Own-price elasticity is usually negative because demand tends to fall as price rises.
- Managers often discuss its absolute value: $|E_d| < 1$ is inelastic and $|E_d| > 1$ is elastic.
- The elasticity may change across prices, customer groups, time periods, and competing alternatives.
- Small-change approximations are less reliable for large price changes.

Managerial caution: Emergency or public-service demand does not imply unlimited ability to charge. Budgets, procurement rules, contracts, and public policy still constrain price.

Back-of-the-Envelope: How Elasticity Affects Revenue

Starting point: Price = HK\$500; volume = 100 passenger-trips; revenue = HK\$50,000. Consider a 20% price reduction.

Approximate $|E_d| = 0.5$

- Approximate volume increase: 10%
- New volume: 110
- New revenue: 400×110
- **HK\$44,000**

Approximate $|E_d| = 2.0$

- Approximate volume increase: 40%
- New volume: 140
- New revenue: 400×140
- **HK\$56,000**

Takeaway: A discount can raise or lower revenue depending on demand response. Profit also depends on capacity and incremental cost.

Illustrative constant-elasticity approximation applied to a relatively large price change.

Demand Heterogeneity

Customers differ in urgency, value of time, budget, reliability needs, accessibility requirements, booking flexibility, and willingness to adopt a new service.

- A single corridor may therefore contain several customer segments with different demand curves.
- Managers may offer different products, booking conditions, service levels, memberships, or time-based prices.
- Segmentation must be operationally feasible, transparent, compatible with applicable law and policy, and acceptable to customers.

Managerial objective: Match the product and price to different needs without assuming that one label perfectly predicts behaviour.

Applied Tool: Revenue by Customer Segment

The Formula

$$\text{Total Revenue} = \sum_{i=1}^n P_i Q_i(P_i)$$

- P_i is the price or product offer for segment i .
- $Q_i(P_i)$ is that segment's expected demand at the offered price.
- Different offers may reflect booking flexibility, priority, refundability, service level, timing, or membership rather than personal characteristics.
- Revenue maximization should be checked against cost, capacity, fairness, transparency, and legal constraints.

Back-of-the-Envelope: Two Product Offers

Scenario: Compare a uniform fare with two differentiated products for an airport corridor.

Uniform Fare: HK\$500

- Flexible-business demand: 4,000
- Advance-purchase demand: 2,000
- **Revenue = HK\$3.00 million**

Differentiated Products

- Flexible fare: HK\$800, demand 3,800
- Restricted advance fare: HK\$300, demand 3,500
- **Revenue = HK\$4.09 million**

Takeaway: Product differentiation can increase revenue under the assumed demand responses. Costs, capacity, cannibalization, fairness, and implementation must also be assessed.

Hypothetical figures for classroom analysis; not a recommendation for pricing based on protected personal characteristics.

What is AiRMOUR?

AiRMOUR was an EU-funded research and innovation project supporting sustainable urban air mobility, with a strong focus on emergency medical services.

- Its guidebook brings together urban design and mobility, aviation safety, public acceptance, and integration-process management.
- It treats public and stakeholder acceptance as part of planning and implementation, not merely as a numerical demand discount.
- Its evidence can help managers identify questions, engagement needs, risks, and scenarios for local analysis.

Source: AiRMOUR, *Guidebook for Urban Air Mobility Integration* (2023).

Public Acceptance: A Hong Kong Application

Acceptance may differ by mission, location, affected group, perceived benefit, noise, privacy, safety, fairness, and trust.

- Emergency medical use may be viewed differently from leisure, passenger, inspection, or commercial delivery use.
- However, “medical” does not imply full support, perfectly inelastic demand, or faster approval.
- Non-medical applications should not be assigned one universal level of opposition.
- Engagement should include users, nearby residents, public agencies, existing transport providers, and other affected stakeholders.

Managerial lesson: Estimate acceptance locally and use scenarios until reliable evidence becomes available.

A Transparent Scenario Adjustment

$$\text{Scenario Demand} = \text{Base Demand} \times \text{Acceptance Adjustment}$$

- The adjustment is an analyst-defined scenario input, not an official AiRMOUR coefficient.
- Use several values, such as 0.6, 0.8, and 1.0, and explain the evidence or judgment behind them.
- Avoid double counting if public acceptance already affects stated mode share, route availability, or generalized cost.
- Over time, replace assumed adjustments with survey, pilot, complaint, booking, and revealed-behaviour evidence.

Back-of-the-Envelope: Acceptance Scenario Analysis

Base forecast: 4,200 potential passenger-trips per day before an acceptance adjustment.

Higher-Acceptance Scenario

- Adjustment: 0.9
- Demand: $4,200 \times 0.9$
- **3,780 passenger-trips/day**

Lower-Acceptance Scenario

- Adjustment: 0.6
- Demand: $4,200 \times 0.6$
- **2,520 passenger-trips/day**

Takeaway: Acceptance assumptions can materially change forecasts. They should be visible, justified, varied, and not presented as measured facts.

Illustrative scenario factors; not coefficients published by AiRMOUR.

Session 3

Scenario Analysis, Uncertainty & Shocks

Concept Primer: Scenario Analysis

A demand forecast should examine plausible changes rather than assume one perfectly stable future.

- **Demand shock:** Fewer customers travel or choose the service.
- **Capacity shock:** Weather, maintenance, staffing, or operating restrictions reduce service availability.
- **Price or cost shock:** Fares, competing-mode prices, energy, insurance, or site costs change.
- **Policy or acceptance change:** New conditions alter routes, operating hours, or customer confidence.

Managerial purpose: Identify which assumptions matter most and prepare responses before committing large investments.

Applied Tool: Revenue Under Demand and Capacity Shocks

Served Quantity

$$Q_{served} = \min\{Q_{demand}, Q_{capacity}\}$$

Revenue

$$R = P \times Q_{served} + \text{Committed Revenue} - \text{Refunds and Credits}$$

- A weather event may reduce capacity rather than customer demand, although repeated unreliability can later affect demand.
- A policy change may affect price, capacity, cost, or demand through different channels.
- Model each channel explicitly rather than applying one unexplained “shock penalty” to revenue.

Back-of-the-Envelope: A Weekly Capacity Shock

Assumptions: Demand and normal capacity = 100 trips/week; price = HK\$500; fixed cost = HK\$40,000/week. Variable cost is omitted for simplicity.

Normal Week

- Served trips: 100
- Revenue: HK\$50,000
- **Margin before variable cost: HK\$10,000**

30% Capacity Loss

- Served trips: 70
- Revenue: HK\$35,000
- **Margin before variable cost: –HK\$5,000**

Takeaway: Possible responses include reserves, flexible capacity, alternative transport, committed revenue, insurance, or revised pricing. Raising the fare to HK\$600 alone would produce HK\$42,000 only if all 70 customers still travelled.

Illustrative assumptions; excludes variable cost, refunds, service credits, and demand response to a price change.

Hong Kong and the GBA: Different Demand Opportunities

Dimension	Hong Kong possibility	Wider GBA possibility
Trip patterns	Dense, complex, constrained	Diverse urban and regional
Customer value	Time-sensitive, specialized	Larger and more varied
Alternatives	Strong public transport	Varies substantially by city
Scale	Limited local geography	Broader network potential
Key question	Value per trip	Volume and network reach

These are hypotheses for analysis, not universal conclusions. Forecasts should use route-specific data on customers, alternatives, access, capacity, and regulation.

In-Class Exercise: Interactive Pricing Simulation

Activity: The Pricing Sweet Spot

- **Format:** Groups of 4–5 (Laptops required)
- **Scenario:** You are managing a HK Airport Shuttle corridor.
- **Task:** Using the provided Excel template:
 1. Set prices for Segment A (Business) and Segment B (Tourists).
 2. Observe how the Elasticity multiplier changes your total volume.
 3. Optimise your prices to achieve maximum total revenue.
- **Timing:** 15 minutes hands-on + 5 minute share-out.

Exercise Template (Visual)

[Distribute digital Excel Spreadsheet to student groups]

The Dashboard View:

Input: Base Price | Executive Discount (%) | Tourist Discount (%)

Output: Executive Volume | Tourist Volume | Total Revenue

Hint: Don't give discounts to executives with inelastic demand!

Key Takeaways

- Demand modelling produces evidence-based scenarios, not exact predictions.
- Distinguish potential passenger demand, available capacity, flights, and passengers actually served.
- Generalized cost compares complete journeys, while mode choice remains heterogeneous and probabilistic.
- Elasticity helps assess price changes, but revenue and profit also depend on capacity and cost.
- Customer segmentation can support differentiated products, subject to transparency, fairness, feasibility, and applicable rules.
- Public acceptance and external shocks should be modelled through explicit, justified assumptions rather than universal multipliers.

Week 4 Assignment Briefing

Demand Forecast Memo and Model (maximum 1,200 words plus spreadsheet)

- **Due:** Beginning of Week 5 (group assignment)
- **Task:** Extend the previous business model with a transparent demand forecast.
- **Requirements:**
 1. Define the relevant trip market, corridor, time period, and customer groups.
 2. Estimate potential demand and mode share using generalized cost or a clearly explained alternative.
 3. Compare forecast demand with available seats and flights.
 4. Estimate price sensitivity for two product or customer segments and state the limitations.
 5. Run one acceptance scenario and one weather or capacity shock.
 6. Recommend a pricing or capacity response and explain its risk.
- **Grading focus:** Economic reasoning (40%), transparent calibration (30%), scenario analysis (20%), professionalism (10%).

Infrastructure Economics and Management

Preview question for next week:

How will these demand forecasts drive exactly where we place million-dollar vertiports in land-scarce Hong Kong?

Thank You + Q&A

Pre-Class Readings Reminder (for Week 5):

- *McKinsey: Perspectives on AAM (Infrastructure sections)*
- *NASA UAM Market Study (Site-selection sensitivities)*

Questions?

(Final 15 minutes reserved for extended Q&A)